FRACTAL ANTENNA DESIGN FOR MULTIBAND APPLICATIONS

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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ABSTRACT

Natural disasters such as earthquakes, tsunamis, and floods cause widespread destruction to both infrastructure and communication networks essential for coordinating rescue operations. When catastrophic events strike, conventional communication systems including cellular networks and base stations collapse due to power outages and physical damage. This breakdown leads to delays in rescue response, poor coordination among relief workers, and preventable loss of human life. Real-world disasters such as the 2011 Japan earthquake and the Haiti earthquake have demonstrated that failure of communication infrastructure is the most critical obstacle in effective disaster management. To overcome this challenge, this project proposes a Disaster Management and Rescue System using Wireless Sensor Networks and Vehicular Ad-hoc Networks. Vibration sensors detect seismic disturbances and transmit alert signals wirelessly through RF modules to a PIC16F877A microcontroller, which relays data to a monitoring PC through an RS-232 interface. When conventional networks fail, VANET nodes formed by vehicles dynamically create a temporary ad-hoc mesh network where every node acts as both transmitter and receiver, enabling communication without any base station or commercial channel. This ensures timely rescue coordination and effective relief distribution during the most critical moments of a disaster.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF ABBREVIATIONS

WSN	Wireless Sensor Network
VANET	Vehicular Ad Hoc Network
MANET	Mobile Ad Hoc Network
ECN-DR	Emergency Communication Network for Disaster Response
PIC	Peripheral Interface Controller
RS-232	Recommended Standard 232
OBC	On Board Computer
TIR	Total Indicator Reading
QoS	Quality of Service
DSRC	Dedicated Short Range Communication
MAC	Medium Access Control
TTL	Transistor-Transistor Logic
CMOS	Complementary Metal Oxide Semiconductor
PMOS	P-type Metal Oxide Semiconductor
EIA	Electronic Industries Alliance
ITU	International Telecommunication Union
ULN	Unipolar Logic Network
GSM	Global System for Mobile Communication
GPS	Global Positioning System
GPR	General Purpose Register
FSR	File Select Register
IoT	Internet of Things

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW:

Fractal antennas, with their unique design that combines multiple scales or shapes (often resembling naturally occurring fractals), can play an impactful role in disaster management due to their versatility and efficiency. Here’s how they can be beneficial:

1. Broadband Performance
2. Compact and Lightweight Design
3. Improved Range and Signal Quality
4. Multi-Functional Capabilities
5. Resilience and Durability

Fractal antennas are more resilient to physical damage compared to conventional antennas due to their simple, scalable design, making them suitable for harsh disaster environments involving high winds, debris, and water exposure.

Applications in Disaster Management:

Emergency Communication Networks: Fractal antennas support temporary communication setups such as mobile command centers, allowing coordination between first responders and government agencies.

Drones for Search and Rescue: Fractal antennas on drones enable better connectivity across diverse frequencies, extending the range and reliability of drone operations.

Sensors and Early Warning Systems: Fractal antennas can be part of IoT-based early warning systems, helping detect environmental changes like landslides or tsunamis and disseminate alerts across a wide area.

Mesh Networks: These antennas help create temporary mesh networks to re-establish communication links for affected communities when traditional networks are down.

Challenges: Fractal antennas come with challenges such as design complexity and ensuring compatibility with existing communication infrastructure in disaster zones. When disasters strike, the 9.0 magnitude earthquake on March 11, 2011 in North-eastern Japan, followed by a 23-meter tsunami and nuclear crisis, identifies key hindrances as follows:

- Paralysis of transportation systems
- Paralysis of communications networks
- Lack of professional disaster response workers
- Dysfunctional administrative command system

Due to poor communication and insufficient information, many precious lives that could have been saved are tragically lost. These communication failures highlight the urgent need for resilient, rapidly deployable antenna systems capable of operating across multiple frequency bands simultaneously. Fractal antennas, with their compact multiband capabilities, offer a promising solution to restore critical communication links swiftly during such catastrophic disaster scenarios, potentially saving countless lives.

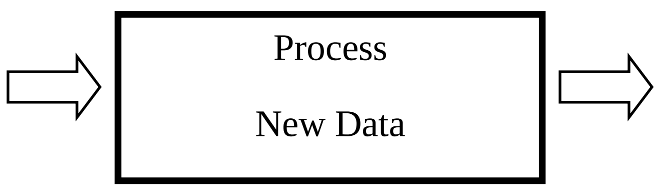
1.2 INTRODUCTION TO EMBEDDED SYSTEM:

An embedded system is any electronic device that incorporates a computer in its implementation. The user is often unaware that a computer is present in the device. Unlike a PC, program code is stored in ROM and not a hard disk drive, and the end user does not develop new software for the embedded device.

1.3 REAL TIME OPERATING SYSTEM:

A real time system must respond to external inputs and produce new outputs within a limited amount of time. Response times that are too long can cause real time systems to fail. A good example is the automobile airbag controller, where the system must deploy the airbag within 10ms of detecting a collision, otherwise the driver will have already impacted the steering wheel before the airbag deploys.

input to response output



Events to Events

System Response Time:

In a soft Real-Time System, critical tasks get priority and normally meet the real-time response constraint. A typical example is a multimedia player, which could occasionally skip a video frame or audio sample without the user noticing, as long as it ran correctly the vast majority of the time. Unlike hard real-time systems, soft real-time systems tolerate minor deadline misses without causing catastrophic failures, making them suitable for consumer applications requiring flexible timing constraints.

1.4 WIRELESS SENSOR NETWORK:

With the popularity of laptops, cell phones, PDAs, and GPS devices in the post-PC era, computing devices have become cheaper, more mobile, and more pervasive, leading to the emergence of wireless sensor networks toward miniaturization of computing devices. A wireless sensor node consists of sensing, computing, communication, and power components. A WSN usually consists of tens to thousands of such nodes communicating through wireless channels for information sharing and cooperative processing. These networks enable real-time environmental monitoring, rapid data collection, and seamless connectivity across vast disaster-affected regions, significantly enhancing situational awareness and emergency response coordination capabilities for rescue teams.

1.5 DISASTER MANAGEMENT AND RESCUE SYSTEM:

Communication systems including cellular networks collapse during disasters, making it essential to have a reliable communication platform to save lives. The suggested system reliably works during catastrophe without using any commercial channels like GSM and GPS. The dedicated system uses sensors that are not affected by interference. A disaster control room monitors the geographical area with an alert system, and relief measures are coordinated through VANET/MANET, forming the Emergency Communication Network for Disaster Response (ECN-DR). This robust, interference-resistant network ensures continuous communication between rescue teams, enabling faster decision-making, efficient resource allocation, and ultimately maximizing the number of lives saved during catastrophic disaster situations.

1.6 OBJECTIVE

To sense and monitor the disaster and to undergo the rescue Operation using Vehicular Ad-hoc networks (VANETs).

1.7 GENERAL BLOCK DIAGRAM

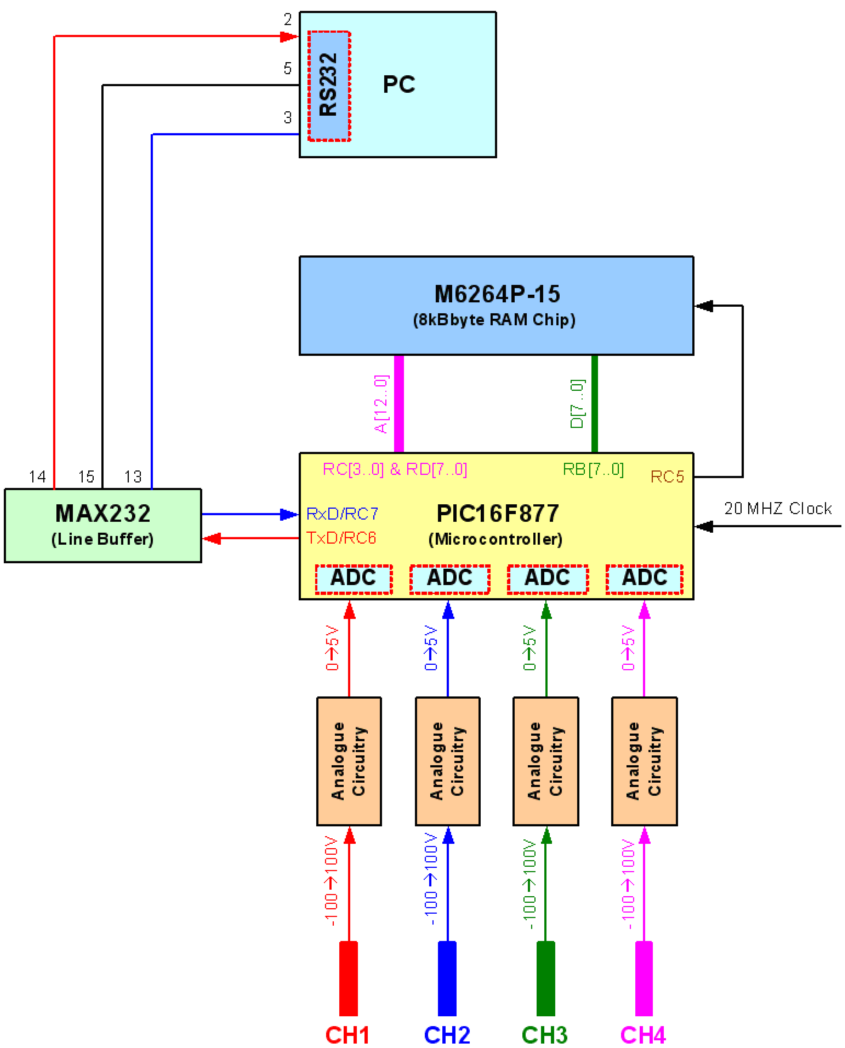


Fig 1.1 Block Diagram

CHAPTER-2

LITERATURE SURVEY

Fractal Antenna-Based Resilient Communication System for Disaster Response and Recovery Networks (2023)

Fractal antennas have emerged as a revolutionary solution for disaster communication due to their multiband operational capabilities and compact design. Traditional single-band antennas fail to maintain reliable connectivity across diverse frequency ranges during catastrophic events. This study proposes a fractal antenna-based communication framework specifically designed for disaster response networks, offering superior signal penetration, extended coverage, and enhanced reliability. The proposed system demonstrates significant improvements in communication resilience, enabling effective coordination among rescue teams and emergency responders during large-scale disaster situations worldwide.

Ubiquitous Robust Communications Emergency Response Using Multi-Operator Heterogeneous Networks (2018)

Disasters have caused extensive loss of lives and severe damages to properties. Robust communications during emergency response are of paramount importance. However, current technologies failed to support mission critical communications due to lack of inter-operability and high vulnerability of centralized infrastructure. Deploying multi-operator heterogeneous networks significantly enhances communication resilience, ensuring seamless interoperability between different network operators and providing uninterrupted, reliable emergency communication services during large-scale catastrophic disaster situations.

A Mobile Ad-hoc Satellite and Wireless Mesh Networking Approach for Public Safety Communications (2016)

Emergency management and disaster recovery systems are of paramount importance worldwide. Public Safety communications infrastructure is a high-priority task due to lack of interoperability between emergency response departments. This work proposes an ad-hoc networking approach combining ad hoc and IPv6 mobility mechanisms for emergency communications in a satellite and wireless mesh scenario. This innovative combination ensures seamless connectivity, extended coverage, and improved network resilience, enabling uninterrupted communication between emergency departments during devastating disaster scenarios.

Framework of an Emergency Management System Using Different Rescue Simulators (2014)

An emergency management system is extremely important during disasters as it provides necessary support to rescue commanders. Different systems have been developed including urban-size disaster simulations and rescue robots for searching victims. This paper presents a framework that links independently developed simulations and connects two disaster and rescue systems. Integrating multiple simulation platforms enhances coordination between rescue teams, improving decision-making accuracy, resource allocation efficiency, and overall disaster response effectiveness during large-scale catastrophic emergency situations. .

CHAPTER-3

FRACTAL ANTENNA DESIGN FOR MUTIBAND APPLICATION

3.1 EXISTING SYSTEM

In the present scenario, after the disaster occurs, information passed to all places through the base station only.

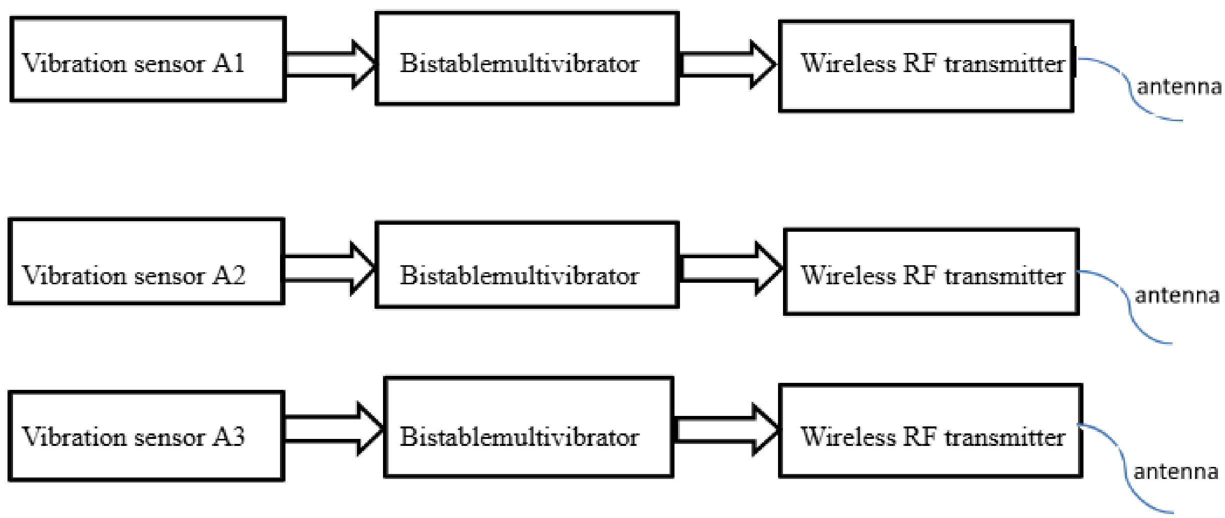
- Creating awareness for improving preparedness among the communities using media the network of the building centre.
- Existing systems heavily rely on centralized infrastructure, making them vulnerable to complete communication failure during large-scale disasters.
- Traditional communication networks lack scalability and flexibility, resulting in significant delays in delivering critical emergency information to affected areas.
- Current systems depend on commercial networks like GSM and GPS, which frequently collapse during catastrophic disaster situations.

3.2 PROPOSED SYSTEM

- When disaster occurs the VANET nodes create temporary Ad-hoc networks.
- In this networks is used to select the nearest VANET
- Every VANET in Ad-hoc networks acts as a transmitter & receiver.
- The information is passed from source to destination through intermediate VANETs without using a base station.

3.3 BLOCK DIAGRAM

MODULE-1



MODULE-2

MODULE-3

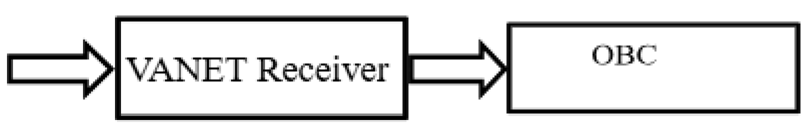


Figure 3.1 Block Diagram

3.4 BLOCK DIAGRAM DESCRIPTION

Peripheral Interface Controller (PIC) is an enhanced version of Microcontroller. It is an embedded controller. The PIC Microcontroller has several advantages over the Microprocessor. The vibration sensor senses the Earth vibration during the earthquake. Schmitt trigger compares and differentiates high and low sensitive earth vibrations. Wireless transmitter and receiver are used to send and receive the information about the high sensitive earthquakes. A microcontroller is used to control the overall application and it gives the digital data via the RS-232 cable. VANET uses vehicles as mobile routers or nodes to create a network with a wide range. OBC is used to display the information received in a VANET.

The GPS module integrated within the VANET node continuously tracks real-time geographical coordinates of vehicles, enabling precise location identification during disaster scenarios. The ZigBee wireless communication module facilitates short-range data transmission between sensor nodes and microcontroller units with minimal power consumption. The temperature and humidity sensors continuously monitor environmental conditions, providing critical data to rescue teams operating in disaster-affected regions. The LCD display unit presents processed sensor data in a human-readable format, allowing field operators to make informed decisions rapidly. The power management unit ensures uninterrupted system operation using solar energy harvesting combined with rechargeable battery backup, maintaining continuous communication even during prolonged power outages in severely affected disaster zones.

CHAPTER 4

HARDWARE IMPLEMENTATION

4.1 OVERALL CIRCUIT AND OPERATION

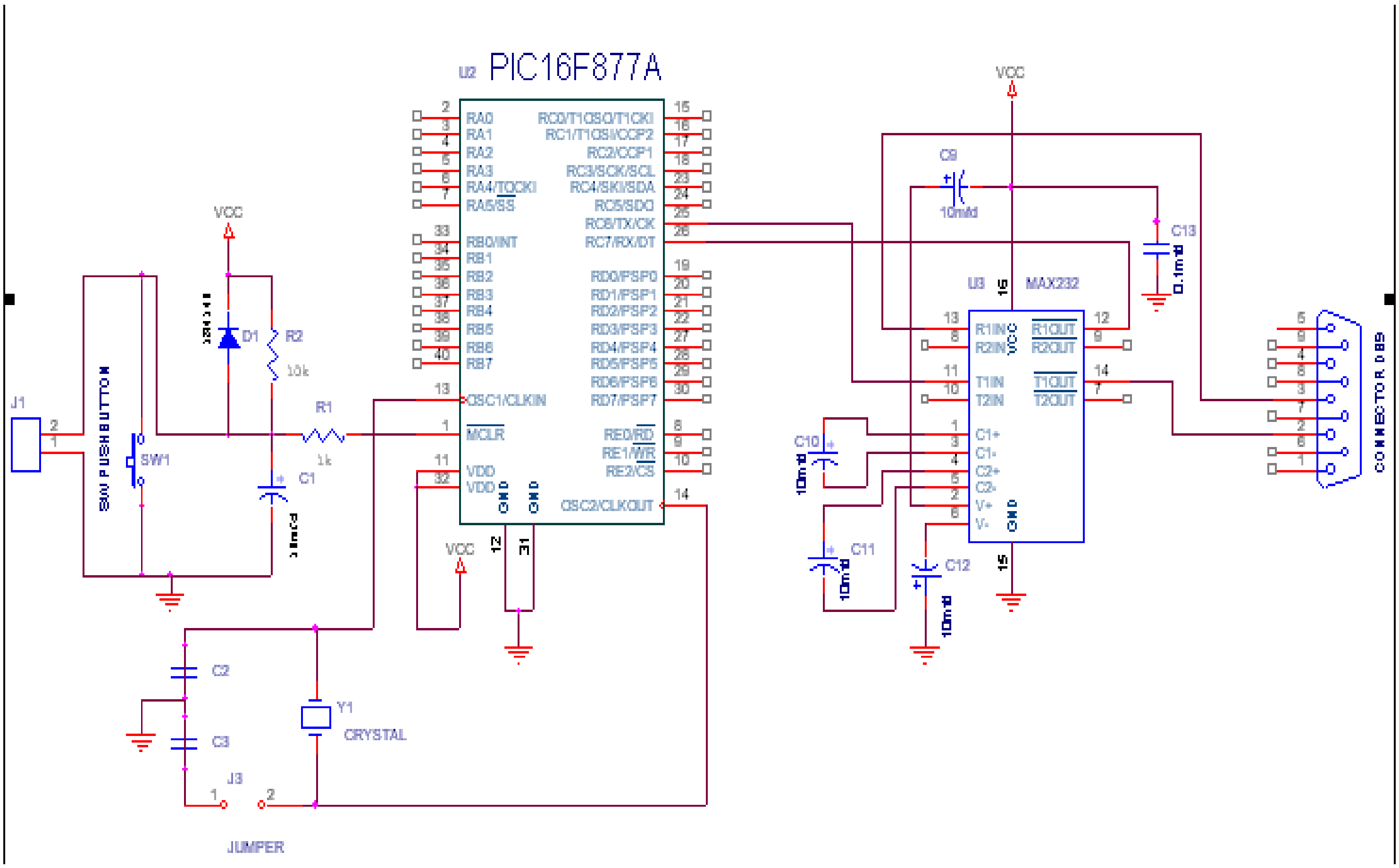


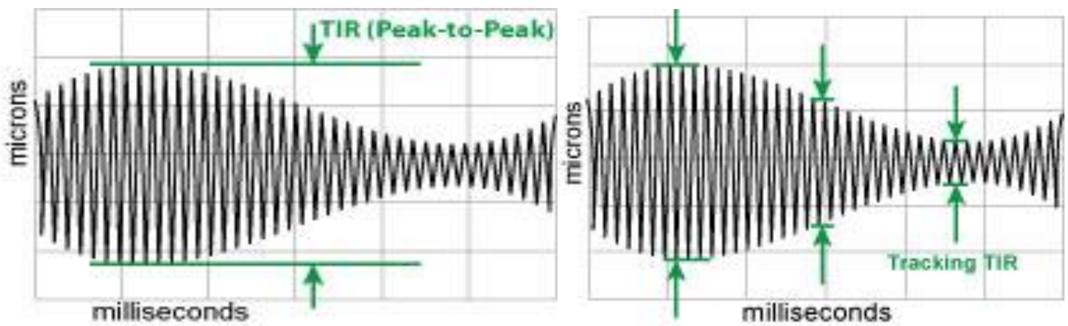
Figure 4.1 Overall Circuit Diagram

4.2 VIBRATION SENSOR

Vibration measurement encompasses components such as displacement, velocity, acceleration, and frequency in both time and frequency domains, where condition monitoring determines kinetic energy and forces acting upon objects, as seen in positional vibration applications like hard-disk drives and machine tools.

MEASURING VIBRATION WITH ACCELEROMETERS:

Accelerometers are small devices installed directly on the surface of the vibrating object, containing a small mass suspended by flexible parts that operate like springs. When moved, the small mass deflects proportionally to the rate of acceleration, which is readily converted to an acceleration value.



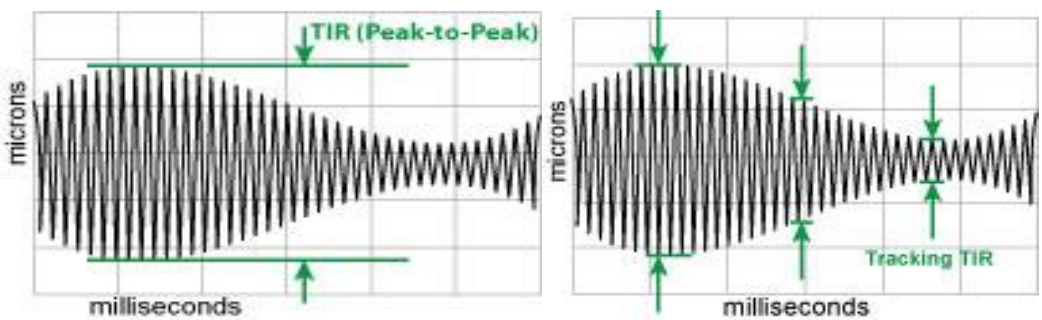
VEHICULAR Ad Hoc NETWORKS:

VANETs have been envisioned for safety, traffic management, and commercial applications. Vehicles can inform others of hazardous road conditions, traffic congestion, or sudden stops. Unicast communication, both single-hop and multi-hop, is one of the key communication paradigms required for VANET applications such as real-time audio/video communication, instant messaging, and coordinated vehicle movement. This paper determines the level of Quality of Service (QoS) available in VANETs using a realistic simulation model implementing PHY and MAC layers of DSRC technology.

VANET ENVIRONMENT CONSIDERED:

A realistic simulation environment is built assuming optimal values of unspecified VANET characteristics. Specified characteristics — including vehicle density, mobility, road topology, signal propagation, and DSRC — currently cannot be influenced by VANET protocol design. For every message between sender and receiver, all available paths are analyzed and the optimal path is selected based on minimum-delay criteria. Only one sender/receiver pair is used per simulation run to avoid interference from other data sources.

The remaining vehicles were distributed randomly. In urban scenarios, the sender/receiver vehicles were initially placed next to each other and moved in a random direction. The remaining vehicles were distributed randomly. The effective range in urban scenarios was up to a maximum of 400 meters, with the vast majority of messages (90+%) received within the 150 meter radius.



INSTANTANEOUS AND TOTAL VIBRATION:

"Total Vibration" can be measured with TIR (peak-to-peak) captures of the vibration signal. Displacement sensors produce real-time outputs observable on an oscilloscope, providing precise vibration data as a function of time or angular location. For applications requiring a simple total vibration number, the MM190 Signal Processing module can derive this measurement using its TIR option, which displays the difference between the most negative and most positive measurements.

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SIMULATION PARAMETERS

Parameter	Value
Number of messages per sender	20 per second
Message size	Normally distributed mean 100 B; std. dev. 15 B
TTL (max. # of hops allowed)	10
Vehicle densities (highway) Vehicle densities (urban) Speeds of vehicles Mean Standard deviation	10, 20, 40 veh/km 10, 50, 100 veh/km^2 ($\approx$ 1, 5, 10 veh/km) Normally distributed highway = 100 km/h urban $\approx$ 40 km/h highway = 25% mean urban = 35% mean
Signal propagation model (highway)	Two ray (based on measurements in [5])
Transmission range (highway)	550 m [5]
Signal propagation model (urban)	Shadowing
Shadowing parameters	$\beta = 2.6$; $\sigma_{dB} = 6$ (based on measurements in [6])
Noise model Transmission rate	Additive channel + thermal noise 6 Mb/s

Table 4.1 Simulation Parameters

In urban VANET scenarios, sender/receiver vehicles moved randomly with an effective communication range of up to 400 meters, where over 90% of messages were successfully received within a 150-meter radius. The remaining vehicles were distributed randomly throughout the simulation environment, and optimal path selection based on minimum-delay criteria ensured reliable and efficient data delivery between source and destination nodes across the network.

4.3 DESIGN OF SYSTEM HARDWARE

The single board is designed around the PIC Micro controller to accomplish the above mentioned system requirements. The hardware section is divided into three subsections as follows:

- Max 232
- Power supply unit
- Pic microcontroller

4.4 MAX 232

.

The Max 232 is a dual RS-232 receiver / transmitter that meets all EIA RS-232C specification while using only a +5V power supply.it has 2 onboard.

The Max 232 is a dual charge pump voltage converter which generates +10V and –10V power supplies from a single 5V power supply. It has four level translators, two of which are RS232 transmitters that convert TTL\ CMOS input levels into + 9V RS232 outputs. The other two level translators are RS232 receivers that convert RS232 inputs to 5V.

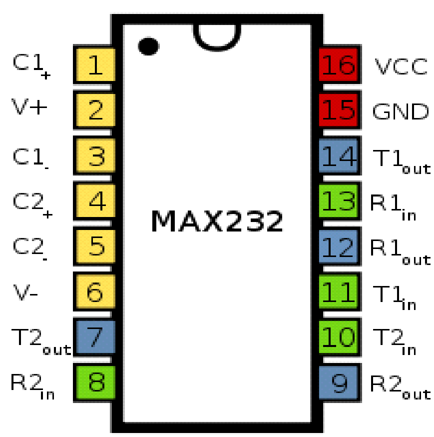


Figure 4.2 Max 232

FEATURES OF MAX 232:

- Meets or Exceeds TIA/EIA-232-F and ITU
- Recommendation V.28
- Operates From a Single 5-V Power Supply
- With 1.0-µF Charge-Pump Capacitors
- Operates Up To 120 kbit/s

POWER SUPPLY SECTION:

The MAX232 power supply section has 2 charge pumps. The first uses external capacitors C1 to double the +5V input to +10V with input impedance of approximately 200Ω. The second charge pump inverts +10V to –10V with an overall output impedance of 45Ω.

TRANSMITTER SECTION:

Each of the two transmitters is a CMOS inverter powered by a +10V internally generated supply, with TTL and CMOS compatible input having a logic threshold of about 26% of VCC. An unused transmitter input can be left unconnected due to an internal 400KΩ pull-up resistor.

RECEIVER SECTION:

The two receivers fully conform to RS232 specifications with input impedance of 3KΩ and switching threshold within +3V of RS232 specification. The MAX232 receivers have VIL of 0.8V and VIH of 2.4V with 0.5V hysteresis for improved noise rejection, ensuring reliable and stable data transmission in noisy industrial and disaster communication environments.

4.5 REGULATORS:

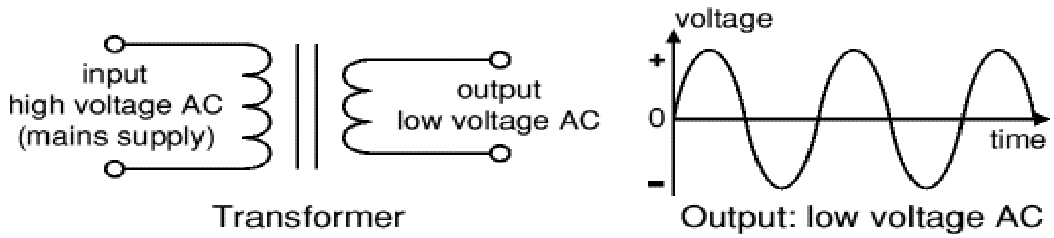
VOLTAGE REGULATORS:

The voltage regulators play an important role in any power supply unit. The primary purpose of a regulator is to aid the rectifier and filter circuit in providing a constant DC voltage to the device.

POWER SUPPLY:

Most power supply circuits are designed to convert high voltage AC mains electricity to a suitable low voltage supply for electronic circuits and other devices. A power supply can be broken down into a series of blocks, each of which performs a particular function.

Transformer



The low voltage AC output is suitable for lamps, heaters, and special AC motors, but is not suitable for electronic circuits unless they include a rectifier and a smoothing capacitor. The output is appropriate for resistive and inductive loads that tolerate the sinusoidal waveform without signal conditioning. Modern electronic circuits, however, require a stable DC supply, so a rectifier must first convert the alternating signal to pulsating DC, followed by a smoothing capacitor to reduce ripple. In high-precision applications, a voltage regulator should also be added to ensure a clean, stable DC rail, as omitting these components risks unreliable operation or permanent damage to sensitive electronics.

REGULATOR

VOLTAGE REGULATOR ICs:

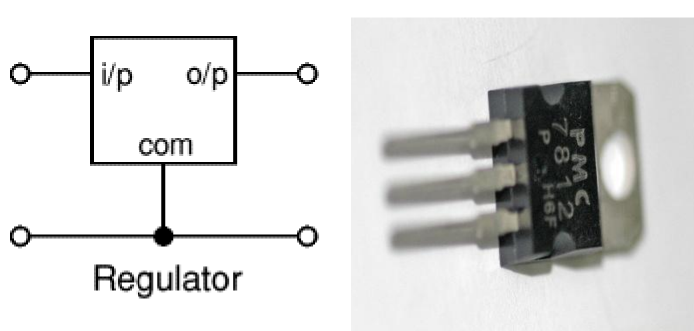


Figure 4.3 Voltage Regulator

Voltage regulator ICs are available with fixed (typically 5, 12 and 15V) or variable output voltages. They are also rated by the maximum current they can pass. Negative voltage regulators are available, mainly for use in dual supplies. Most regulators include some automatic protection from excessive current (overload protection) and overheating.

4.6 RELAYS:

A relay is an electrical switch that opens and closes under control of another electrical circuit. In the original form, the switch is operated by an electromagnet to open or close one or many sets of contacts.

RELAY FEATURES:

- Very low power consumption typically 360mW
- 24V DC or 110V AC
- Flux tight construction
- Small size, less than 0.53 cubic inches

4.7 RELAY DRIVER ULN2003

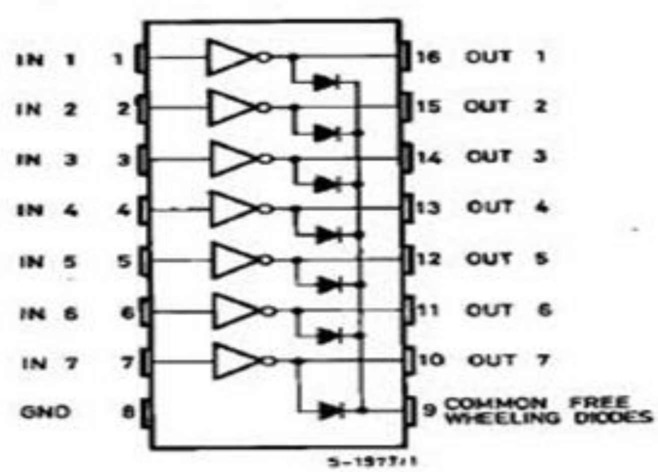


Figure 4.4 Relay Driver

ULN2003 DARLINGTON TRANSISTOR ARRAYS:

The ULN2003A/L operates with 5V TTL/CMOS inputs handling interface needs beyond standard logic buffers, while the ULN2004A/L supports 6 to 15V CMOS or PMOS logic outputs directly.

FEATURES:

- Transient-Protected Outputs
- Dual In-Line Plastic Package or Small-Outline IC Package

4.8 RELAY DRIVING CIRCUIT

Relay driving circuit comprises of ULN 2003 to which 4 relays are connected. Below is the relay driving circuit and its operation.

RELAY CIRCUIT OPERATION

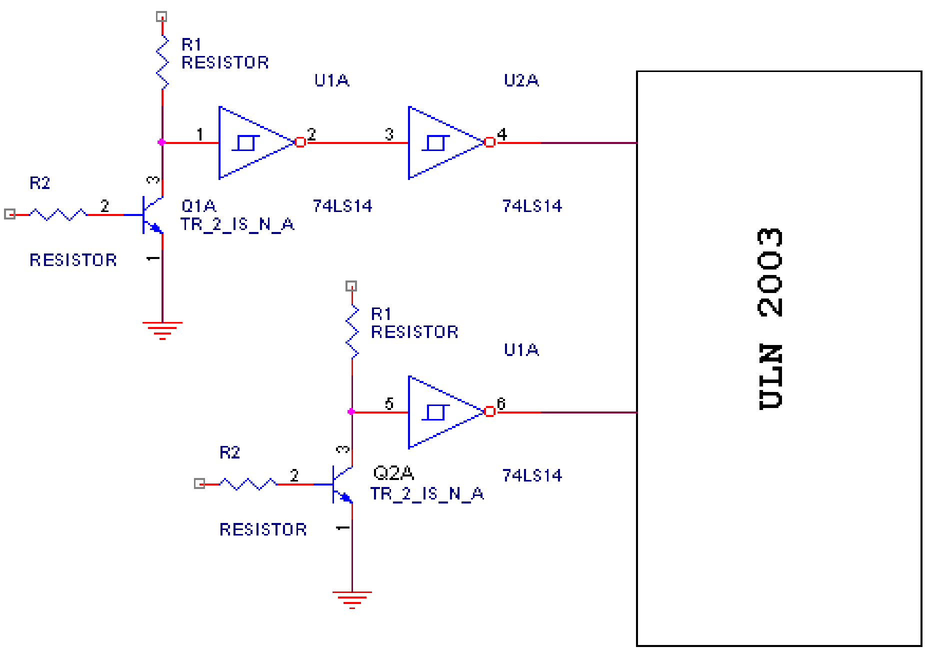


Figure 4.5 Relay Driving Circuit

In the transmitter section the output of the voltage and current sensing circuit is given as input to the relay circuit. When the voltage sensing circuit senses the low voltage in area1 it is given as input to the Q1A. So the output is high. Then this high output double inverted using Schmitt trigger IC. Then the output of the Schmitt trigger IC (PIN 4) given as input to the relay driver IC ULN2003. The IC operates the corresponding relay, then the relay output is connected to the FM transmitter. The corresponding signal transmitted.

RS-232 CONVERTER:

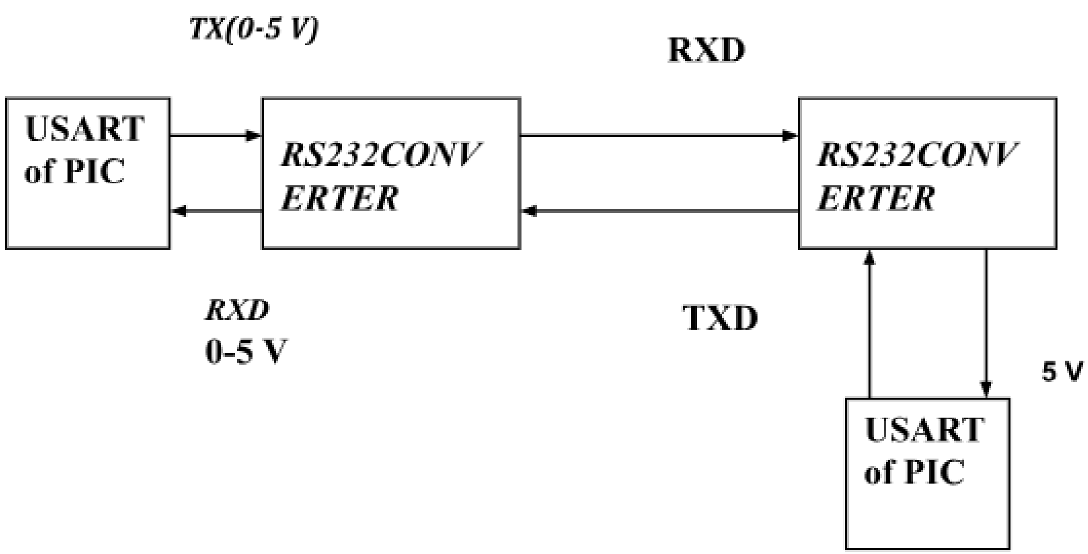


Figure 4.6 RS-232 Converter

RS – 232 The RS-232 converter interfaces the PC with the PIC microcontroller by converting parallel data to serial form for transmission. It scales the 5V PIC output to ±10V (20V) for PC communication, and steps down the incoming 20V PC signal back to 5V for the microcontroller.

4.9 RS 232

Serial communication is the transmission or reception of data one bit at a time, where the serial port and UART convert each byte into a sequential bit stream of ones and zeroes, operating in simplex, half-duplex, or full-duplex modes, making it widely adopted in embedded systems and peripheral interfacing due to its simplicity and reduced wiring complexity compared to parallel communication. Additionally, serial communication supports various standard protocols such as RS-232, SPI, and I2C, each offering different data rates, voltage levels, and communication distances to suit a wide range of industrial and consumer applications. Error detection mechanisms such as parity bits, checksums, and stop bits are commonly incorporated to ensure data integrity during transmission. As systems demand higher reliability and longer communication distances, protocol selection and proper line termination become critical design considerations in both industrial and consumer embedded applications.

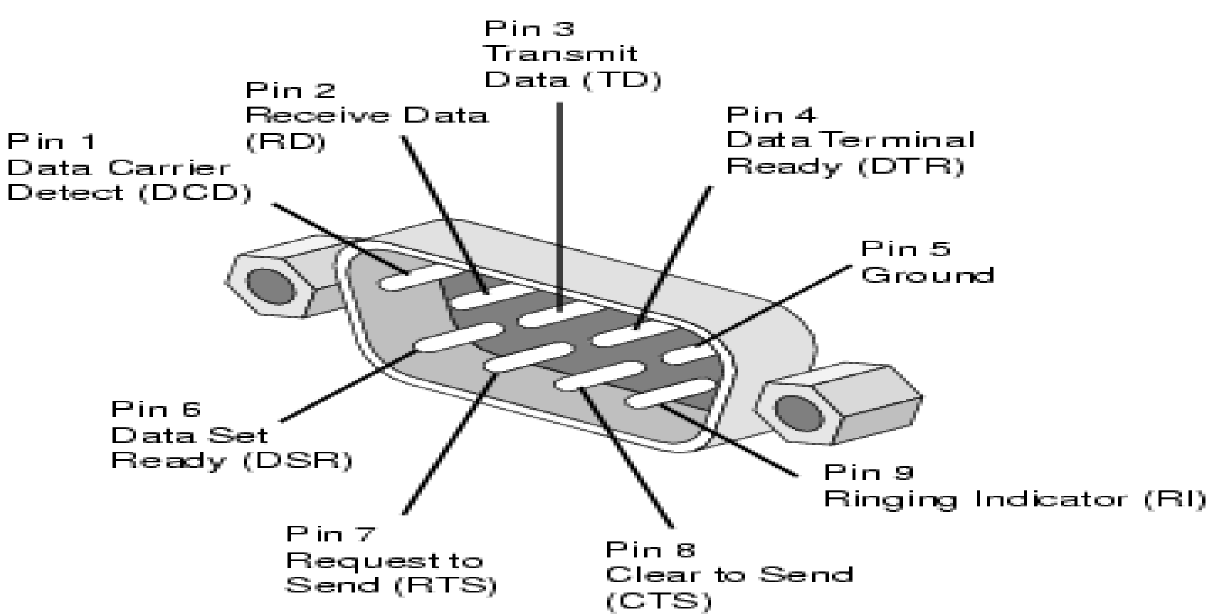


Figure 4.7 RS-232 Pin

CHAPTER-5
MICROCONTROLLER

5.1 INTRODUCTION TO MICROCONTROLLER

An Embedded system is a special purpose computer controlled electro-mechanical system in which the computer is completely encapsulated by the device it controls. An embedded system has specific requirements and performs pre-defined tasks, unlike a general-purpose personal computer. These systems are designed to operate with minimal human intervention, making them highly reliable and efficient for dedicated applications. They are widely used in consumer electronics, automotive systems, medical devices, and industrial control systems. Embedded systems typically combine hardware and software components, where the software is permanently stored in ROM or Flash memory to ensure consistent performance. Due to their task-specific nature, they are optimized for low power consumption, compact size, and real-time responsiveness, making them ideal for resource-constrained environments.

Figure 5.1 Microcontroller (PIC16F877A)

5.2 PIN CONFIGURATION

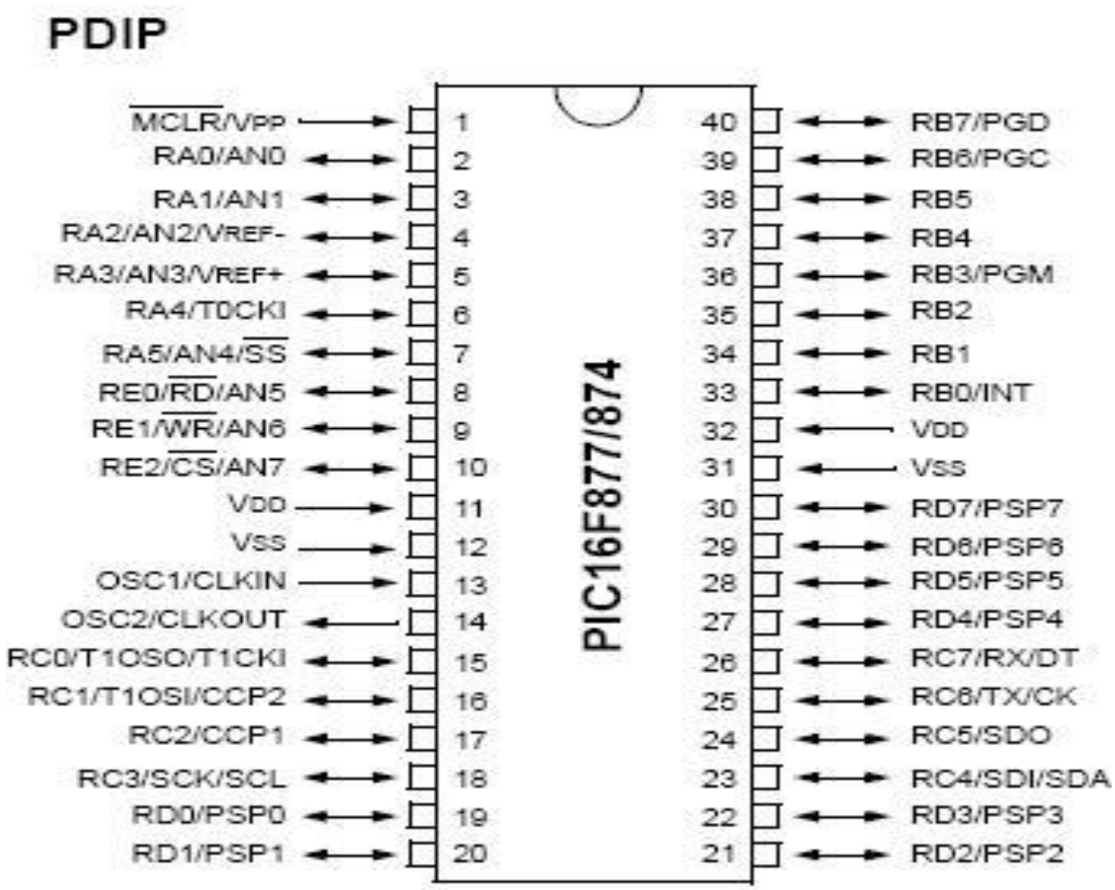


Figure 5.2 Pin Configuration (PIC16F877A)

5.3 PROGRAM MEMORY ORGANISATION

The The PIC16F87XA devices have a 13-bit program counter capable of addressing an 8K word x 14 bit program memory space, where the Flash memory allows electrical erasing and reprogramming, while the 14-bit wide instruction set enables efficient single-cycle instruction execution for high-performance embedded applications.

DIAGRAM:

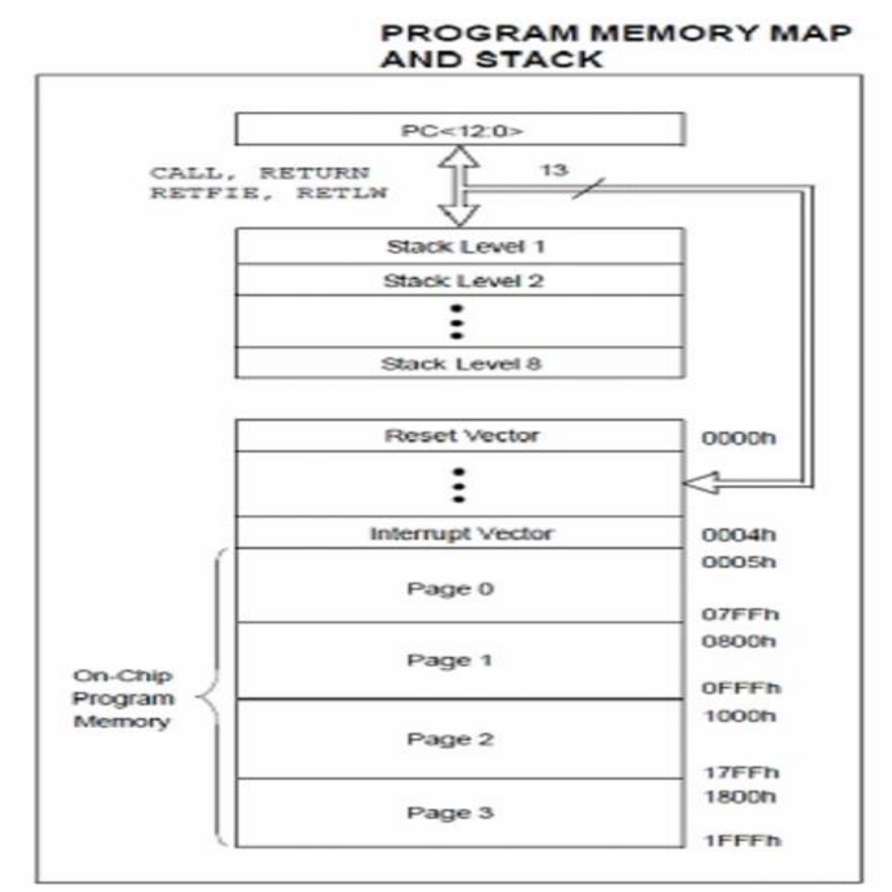


Figure 5.3 Memory Organisation of PIC16F877AA

5.4 I/O PORTS

PORT A: PORT A is a 6-bit wide, bidirectional port. Setting a TRIS A bit will make the corresponding PORT A pin an input in High-Impedance mode.

PORT B: PORT B is an 8-bit wide, bidirectional port. Setting a TRIS B bit will make the corresponding PORT B pin an input in High-Impedance mode.

PORT C: PORT C is an 8-bit wide, bidirectional port. PORT C pins have Schmitt Trigger input buffers.

5.5 MEMORY ORGANIZATION OF PIC16F877A

The memory of a PIC16F877 chip is divided into 3 sections. Program memory contains the programs written by the user. The program counter executes stored commands one by one with a 13-bit wide program counter.

GENERAL PURPOSE REGISTERS (GPR):

GPR is a small amount of storage accessible more quickly than any other memory. These register files can be accessed either directly or indirectly through the File Select Register (FSR). The direct addressing mode uses the address embedded within the instruction itself, while indirect addressing allows dynamic memory access by loading the desired register address into the FSR, providing greater flexibility in handling data during program execution.

Device	Program Memory		Data SRAM (Bytes)	EEPROM (Bytes)	I/O	10-bit A/D (ch)	CCP (PWM)	MSSP		USART	Timers 8/16-bit	Comparators
	Bytes	# Single Word Instructions						SPI	Master I^2C			
PIC16F873A	7.2K	4096	192	128	22	5	2	Yes	Yes	Yes	2/1	2
PIC16F874A	7.2K	4096	192	128	33	8	2	Yes	Yes	Yes	2/1	2
PIC16F876A	14.3K	8192	368	256	22	5	2	Yes	Yes	Yes	2/1	2
PIC16F877A	14.3K	8192	368	256	33	8	2	Yes	Yes	Yes	2/1	2

Table 5.1 General Purpose Registers

SOFTWARE IMPLEMENTATION

Step 1: Openmicrosoft visual basic

Step 2: Create a new vb file by clicking on a new project.

Step 3: Click standard exe on the new project window.

Step 4: Open a new form and assemble the buttons

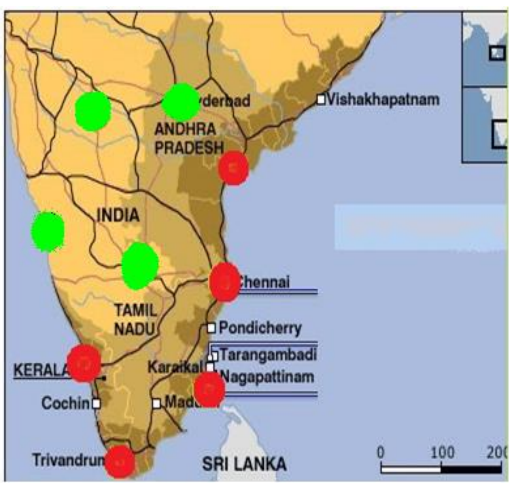
Step 5: Click and drag the tools to be added on to form.

Properties of each tool can be changed with the help of the properties box.

Double click on the command button to get the following page.

Step 7

Final simulation output.



CHAPTER-6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION

When stricken by a catastrophic natural disaster, many communication systems crashed, including cellular networks. The loss of the communication system may have a catastrophic consequence. From the Chi-Chi Earthquake and 88 Flood, we learn that power outage and backhaul link breakage were the two common problems that crushed base stations. It is difficult to enhance the availability of power lines and backhauls since they are highly dependent on the robustness of roads and bridges.

6.2 FUTURE ENHANCEMENT

The future of ad-hoc networks is appealing, giving the vision of anytime, anywhere and cheap communications. Before those scenarios come true, a huge amount of work is to be done in both research and implementation. At present, the general trend in MANET is toward mesh architecture and large scale. Improvement in bandwidth and capacity is required, which implies the need for higher frequency and better spatial spectral reuse. Propagation, spectral reuse, and energy issues support a shift away from a single long wireless link to a mesh of short links as in ad-hoc networks. Large scale ad hoc networks are another challenging issue in the near future. As involvement grows, the nodes in ad-hoc networks will be smaller, cheaper, more capable, and come in all forms. In all, the widespread deployment of ad-hoc networks is still a year away, but research will continue being very active and imaginative.

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